

Advancing the Understanding of Fixed Point Iterations in Deep Neural Networks: A Detailed Analytical Study

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Abstract

Recent empirical studies have identified fixed point iteration phenomena in deep neural networks, where the hidden state tends to stabilize after several layers, showing minimal change in subsequent layers. This observation has spurred the development of practical methodologies, such as accelerating inference by bypassing certain layers once the hidden state stabilizes, selectively fine-tuning layers to modify the iteration process, and implementing loops of specific layers to maintain fixed point iterations. Despite these advancements, the understanding of fixed point iterations remains superficial, particularly in high-dimensional spaces, due to the inadequacy of current analytical tools. In this study, we conduct a detailed analysis of fixed point iterations in a vector-valued function modeled by neural networks. We establish a sufficient condition for the existence of multiple fixed points of looped neural networks based on varying input regions. Additionally, we expand our examination to include a robust version of fixed point iterations. To demonstrate the effectiveness and insights provided by our approach, we provide case studies that looped neural networks may exist 2^d number of robust fixed points under exponentiation or polynomial activation functions, where d is the feature dimension. Furthermore, our preliminary empirical results support our theoretical findings. Our methodology enriches the toolkit available for analyzing fixed point iterations of deep neural networks and may enhance our comprehension of neural network mechanisms.

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1 INTRODUCTION

Deep neural networks have achieved remarkable success and are widely employed in various applications, including ChatGPT [Ope23], face recognition [WD21], and personalized recommendation systems [DS20], among others. These networks typically consist of numerous hidden layers; for instance, Residual Networks (ResNets) [HZRS16] can contain over 1,000 layers.

Recent empirical studies reveal that, despite the numerous layers in deep neural networks, certain operational phases exist where adjacent layers may perform identical operations [MBAB22, SMN⁺24]. Consequently, we can focus on modifying specific associated layers during inference to enhance performance. Furthermore, the hidden states tend to stabilize after several adjacent layers, resulting in minimal changes in subsequent layers, allowing us to skip certain layers during inference [ESL⁺24]. Additional research indicates that a looped transformer—where its output is fed back into itself iteratively—exhibits expressive capabilities comparable to programmable computers [GRS⁺23] and is more effective at learning algorithms [YLN⁺23]. Other studies have also examined the convergence of deep neural networks when the weights across different layers are nearly identical, with only minor perturbations [XZ22, XZ24]. Collectively, these findings suggest the relevance of fixed-point iteration (Definition 3.1). Employing fixed-point methods in deep or looped neural networks may offer several advantages, such as reducing the number of parameters and dynamically adjusting runtime based on the complexity of the problem.

Despite these advancements, our understanding of fixed-point iterations remains limited, especially in high-dimensional spaces, due to the limitations of existing analytical tools. It remains challenging to determine when a neural network can effectively approximate a fixed-point solution and how many layers or iterations are required to ensure a good outcome.

Thus, it is natural to ask the following question:

How shall we analytically study fixed point iterations in deep neural networks?

In this study, we conduct a detailed analysis of fixed point iterations in a vector-valued function, $\mathbb{R}^d \rightarrow \mathbb{R}^d$ where d is the hidden feature size, modeled by neural networks. We establish a general theorem (Theorem 4.1) to describe a sufficient condition for the existence of multiple fixed points of looped neural networks (Definition 3.5) based on varying input regions. Then, we expand our examination by introducing noise during each fixed point iteration and show that the fixed point iteration process is robust under noise (Theorem 4.2). It represents deep neural networks with residue connection [HZRS16], where after each layer, the hidden states are only perturbed slightly. Finally, we demonstrate the effectiveness of our approach by studying looped neural networks under polynomial and exponential activation functions (Theorem 5.1 and Theorem 5.2). We show that the looped neural networks may exist 2^d number of robust fixed points. Recall that the previous tools can only handle single fixed point analysis, while our analysis can be applied to more practical cases. Furthermore, our preliminary empirical results support our theoretical findings (Section 6). Our methodology enriches the toolkit available for analyzing fixed point iterations of vector-valued functions and may help us better understand neural network mechanisms.

Our contributions:

- We study the fixed point iteration in looped neural networks and provide a general theorem (Theorem 4.1) to describe a sufficient condition for the existence of multiple fixed points.
- We also establish a robust version of fixed point iterations with noise perturbation (Theorem 4.2).

- We provide two case studies, where looped neural networks may have 2^d number of robust fixed points, which demonstrates the effectiveness of our approach (Theorem 5.1 and Theorem 5.2).
- Our preliminary empirical results validate our theoretical findings (Section 6).

Roadmap. Our paper is organized as follows. In Section 2, we review related literature. In Section 3, we present the preliminary of our notations, Banach fixed point theorem and our definition of Looped Neural Network. In Section 4, we outline the main results of this work. In Section 5, We present case analysis results of fixed-point iterations for neural networks using two types of activation functions. In Section 6, we present the experiment result of this work. In Section 7, we conclude our paper.

2 RELATED WORK

In Section 2.1, we introduce fixed point theory with a focus on the Banach fixed point theorem. In Section 2.2, we present some work on incorporating looped structures into neural networks. In Section 2.3, we introduce some works that utilize the properties of fixed point iterations in neural network computations.

2.1 Fixed Point Iteration Methods

In numerical analysis, fixed point iteration methods [AMO01, Ist01, GD03, KSR15] use the concept of fixed points to compute the solution of a given equation in a repetitive manner. Many works have focused on the convergence properties of fixed point iteration methods. For example, Banach fixed point theorem [AH09] gives a sufficient condition under which a unique fixed point exists and is approachable via iterative methods. Although there are other fixed-point theorems, Banach fixed point theorem, in particular, is useful because it provides a clear criterion for fixed points using contraction mappings. If a function is a contraction, the theorem guarantees the existence and uniqueness of a fixed point, making it easier to work with than other fixed-point theorems that may have more complex conditions.

Recent works focus on employing various methods to accelerate the convergence of fixed-point iterations. For instance, [ZAL11] proposed a Quasi-Newton method for accelerating fixed-point iterations by approximating the Jacobian in Newton’s method, enabling efficient root-finding for the function $g(x) = x - f(x)$. [WN11] presents Anderson acceleration, an underutilized method for enhancing fixed-point iterations. After this work, [ZOB20] introduces a globally convergent variant of type-I Anderson acceleration for non-smooth fixed-point problems, improving terminal convergence of first-order algorithms.

2.2 Looped Neural Networks

Looped Neural Networks are a paradigm in deep learning that aims to address certain limitations of traditional feedforward architectures. The addition of loopy structures to traditional neural networks has already received extensive research. For example, [CSW16] introduces looped Convolutional Neural Network which unrolls over multiple time steps and demonstrate that these networks outperform deep feedforward networks on some image datasets.

Transformers [VSP⁺17, CSY23, LSS⁺24b] have become the preferred model of choice in natural language processing (NLP) and other domains that require sequence-to-sequence modeling. To understand why Transformers excel at iterative inference while lacking an iterative structure, [GRS⁺23] proposes a Looped Transformer and has shown that transformer networks can be used

as universal computers by programming them with specific weights and placing them in a loop. [GSR⁺24] investigates the learnability of linear looped Transformers for linear regression, showing they can converge to algorithmic solutions via multi-step preconditioned gradient descent with adaptive preconditioners. [GZX⁺24, FDRL24, XS24, CLL⁺24, LSS⁺24a] argued that the Looped Transformer could achieve significantly higher algorithmic representation capabilities and good generalization ability while using the same number of parameters compared to the standard Transformer.

2.3 Neural Networks as Fixed Point Iterations

To better understand the convergence property and stability of neural networks, many works investigate neural networks as fixed-point iteration processes. The research in this area can be traced back to [HO97]. This work demonstrates how a neural network learning rule can be converted into a fixed-point iteration, resulting in a simple, parameter-free algorithm that converges quickly to the optimal solution allowed by the data. Recently, many researchers have found that the hidden layers of many deep networks converge to a fixed point (a stable state). Based on this, treating neural networks as fixed points has received extensive research. For instance, [BKK19] proposes Deep Equilibrium Model (DEQ), a novel approach to modeling sequential data by directly solving for fixed points. Instead of gradually approximating this fixed point through iterations, DEQ uses a black-box root-finding method to solve it directly, bypassing intermediate steps. [YLNP23] proposes a new training methodology for looped transformers to effectively emulate iterative algorithms, optimizing convergence by adjusting input injection and loop iterations while using significantly fewer parameters than standard transformers. It demonstrates that the looped Transformer is more effective at learning learning algorithms such as in-context learning. There are many other works used fixed point iterations in deep learning [JLC21, HGM⁺21, CMK15].

We refer the readers to some other related works [QGT⁺19, RSZ22, GSX23, RHX⁺23, HWL⁺24, DSX23, DSXY23, QSZZ23, QSY23, AS24b, AS24a, WHL⁺24, ASZZ24, LSSY24, LLSS24, SYZ24, SZZ24]

3 PRELIMINARY

In this section, we introduce some definitions that will be used throughout the paper in Section 3.1. Then we briefly review the fixed point method and Banach fixed-point theorem in Section 3.2. Finally, we define the looped neural network in Section 3.3, which is the primary focus of our work.

3.1 Notations

For a vector $v \in \mathbb{R}^d$, we use $\|v\|_1$, $\|v\|_2$, and $\|v\|_\infty$ to denote the ℓ_1 -norm, ℓ_2 -norm, and ℓ_∞ -norm of v , respectively. For two vector $u, v \in \mathbb{R}^d$, we use $\langle u, v \rangle$ to denote the standard inner product of u and v . We use $\mathbf{1}_d$ to denote a vector whose elements are all 1.

3.2 Fixed Point Methods

In this section, we introduce the concept of the fixed-point method and the well-known Banach fixed-point theorem. Here, we only deal with the case where the space is $\mathbb{R}^d$. In Appendix A, we state the original definitions and theorems in the context of Banach spaces.

We first introduce the fixed point iteration problem.